

UPERNAVIK, Greenland

WMO: 042110

Lat: 72.783N

Lon: 56.133W

Elev: 423

StdP: 14.47

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-22.7	-19.5	-27.4	1.2	-22.5	-24.2	1.5	-19.1	43.5	21.2	31.6	11.1	5.2	60	0.956

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	57.2	45.9	54.8	44.8	52.3	43.7	47.1	54.2	45.9	52.2	44.8	50.6	6.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.0	43.2	46.4	42.6	40.8	45.5	41.1	38.6	44.9	18.8	54.5	18.2	52.6	17.6	50.9	54.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.5	28.5	22.6	DB	-20.6	61.2	4.4	3.2	-23.8	63.5	-26.4	65.4	-28.9	67.2	-32.1	69.5	(4)
(5)			WB	-21.0	50.0	4.5	2.5	-24.2	51.8	-26.9	53.3	-29.5	54.6	-32.8	56.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	4.0	-0.3	-0.9	11.6	26.4	38.7	44.9	43.7	35.4	26.2	19.4	12.5		
	DBStd	17.96	11.36	12.79	11.85	9.53	7.45	6.53	5.70	4.60	4.48	5.73	6.64	8.60		
	HDD50	10274	1426	1407	1579	1152	731	343	178	203	437	737	918	1163		
	HDD65	15720	1891	1827	2044	1602	1196	790	624	661	887	1202	1368	1628		
	CDD50	30	0	0	0	0	0	4	19	7	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.0	9.4	7.7	7.6	8.7	8.0	8.0	7.4	7.7	8.7	10.9	12.3	11.6		
	PrecAvg	9.5	0.4	0.4	0.3	0.4	0.4	0.6	1.0	1.2	1.4	1.2	1.3	0.8		
	PrecMax	13.8	2.1	1.6	0.7	1.1	1.4	1.9	2.9	3.8	4.0	2.8	4.5	1.7		
(15) Precipitation	PrecMin	4.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0		
	PrecStd	2.6	0.4	0.4	0.2	0.3	0.3	0.5	0.8	0.9	1.1	0.7	1.0	0.5		
	DB	35.1	36.3	37.0	40.8	50.2	58.3	61.8	58.8	48.0	42.6	41.0	36.0			
	MCWB	31.0	30.4	32.6	35.6	41.4	46.0	48.1	47.1	39.6	35.7	36.8	31.4			
	2% DB	29.9	29.9	29.9	33.6	44.6	54.7	58.2	55.6	44.7	39.2	35.0	31.6			
(21) Mean Coincident Wet Bulb Temperatures	MCWB	26.7	26.0	26.1	29.5	37.2	44.4	46.8	44.9	37.6	33.9	31.6	27.8			
	5% DB	24.5	24.3	22.8	28.9	40.6	51.7	55.7	53.2	42.7	36.5	32.1	27.9			
	MCWB	21.3	21.9	19.8	25.7	35.4	42.6	45.5	44.2	37.4	32.7	28.7	25.5			
	10% DB	19.1	18.3	15.2	24.4	36.1	48.8	53.4	50.6	41.0	33.7	29.2	23.5			
	MCWB	16.9	16.6	13.6	22.5	32.9	40.9	44.2	43.2	36.8	31.2	26.9	21.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	32.3	30.3	32.4	35.7	42.2	46.8	49.4	49.0	41.2	37.1	37.8	32.5			
	MCDB	33.5	35.1	34.6	40.4	48.7	55.9	58.5	55.5	44.3	40.4	40.4	36.8			
	2% WB	27.1	25.9	24.8	29.2	38.0	44.9	47.7	46.7	39.9	34.4	32.3	28.6			
	MCDB	29.7	29.1	27.9	32.9	42.8	53.5	55.5	52.1	42.8	37.4	34.5	30.9			
	5% WB	21.9	20.4	18.4	25.9	35.6	43.2	46.5	45.5	38.8	32.8	29.6	25.6			
(31) Mean Daily Temperature Range	MCDB	24.8	22.5	20.7	28.9	40.1	50.6	53.5	50.4	41.4	35.4	31.7	28.4			
	10% WB	17.5	15.7	13.0	22.3	33.1	41.4	45.4	44.3	37.4	31.1	26.9	21.9			
	MCDB	19.4	17.6	14.3	24.5	36.0	47.5	51.7	48.9	40.2	33.3	29.5	23.9			
	5% DB	12.8	14.6	13.3	13.5	13.3	13.1	12.4	11.1	7.0	7.8	8.8	9.6			
	MCWBR	10.6	11.0	10.6	10.3	7.9	6.6	6.2	5.9	4.7	5.6	7.2	7.5			
(37) Clear-Sky Solar Irradiance	5% WB	12.7	13.5	12.5	13.5	12.4	13.1	11.3	9.7	6.1	6.5	9.4	9.8			
	MCWBR	10.8	11.4	10.9	10.8	8.0	7.1	6.3	5.7	4.6	5.4	8.1	8.0			
	taub	N/A	0.158	0.236	0.282	0.324	0.324	0.316	0.309	0.278	0.356	N/A	N/A			
	taud	N/A	1.941	2.160	2.126	2.196	2.455	2.550	2.555	2.489	2.938	N/A	N/A			
	Ebn at Noon	0	152	234	261	265	272	271	258	226	130	0	0			
(42) All-Sky Solar Radiation	Edh at Noon	0	9	20	32	35	28	25	21	16	7	0	0			
	RadAvg	0	67	535	1339	1922	2056	1756	1085	523	119	0	0			
(44) All-Sky Solar Radiation	RadStd	0	6	26	83	100	162	155	120	34	11	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page